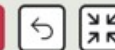


PROTEINLOOP tank

Off-grid AI for living protein systems

Gemma 4 | DECT NR+ | Deterministic safety

⚡ Simulate water emergency



AGENTIC AI CONTROL

Gemma configured

Live recovery mission

Mission

Recover water quality

Recover water quality

Reduce ammonia and restore dissolved oxygen while protecting fish and prawn survival.



5-agent team



Safety verifier



Human approval



Run agent team

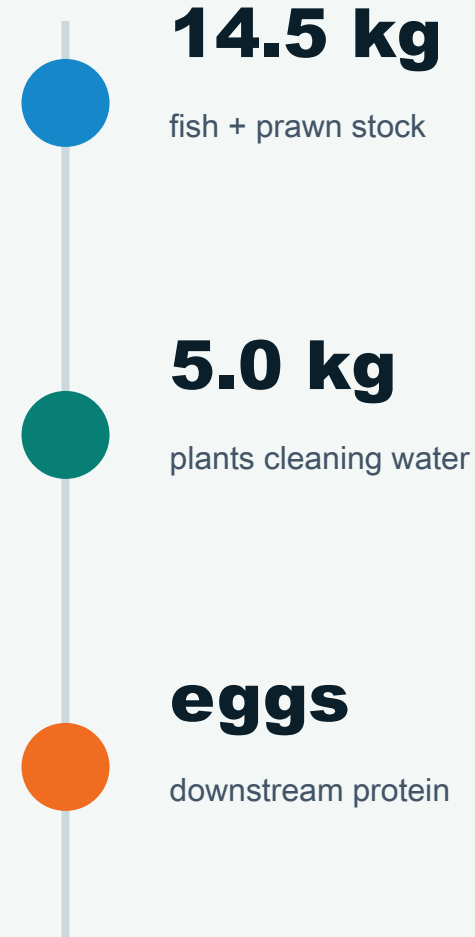
ONE LOOP, SHARED RISK

A chemistry failure does not stop at the vegetables.

Fish and prawns breathe the water. Plants clean it. Duckweed stores feed. Eggs depend on what survives upstream.

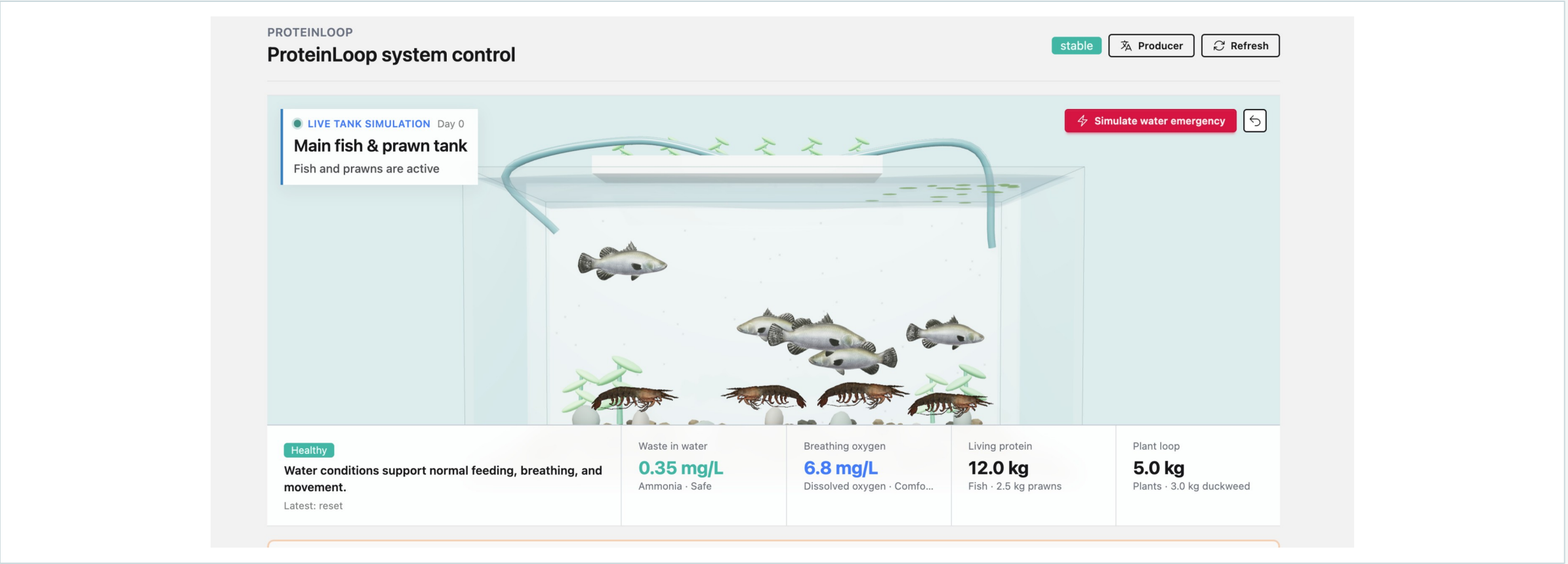
One ammonia spike

can threaten every protein output in the loop.



A living system people can understand in seconds.

The deployed operator view connects animal behavior, plain-language chemistry, biomass, and action in one scene.



SEE THE ANIMALS

UNDERSTAND THE WATER

ACT WITH EVIDENCE

Real deployed UI at proteinloop.dev-vb.lat; the tank is rendered live, not a looping video.

The recovery is visible while it happens.

1 Observe

2 4 specialists

3 Supervisor

4 Verify

AGENTIC AI CONTROL

Gamma configured

Live recovery mission

5 Measure

Mission

Recover water quality

Recover water quality

Reduce ammonia and restore dissolved oxygen while protecting fish and prawn survival.



5-agent team



Safety verifier



Human approval



Run agent team

Structured events and tool outcomes are visible; private model chain-of-thought is not presented as evidence.

Needs attention

Waste in water

0.75 mg/l

Breathing oxygen

0.0 mg/l

Living protein

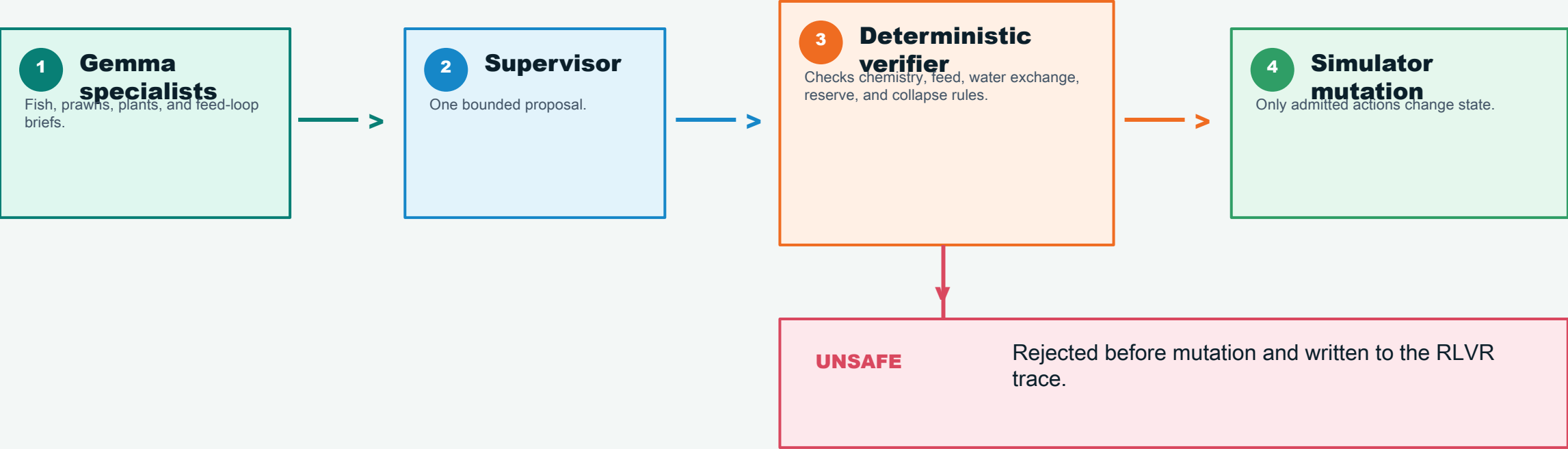
10.0 mg/l

Plant loop

5.00 mg/l

The model can recommend. It cannot mutate.

Authority moves left to right. Every connector ends at an explicit decision boundary.



Safety is demonstrated, not narrated.

The simulator is simultaneously the product demo, the policy evaluator, and the RLVR source of truth.

180

Python tests

simulator, verifier, API, evidence

129

Phoenix tests

Sagents, LiveView, producer control

+463

reward delta

verified policy versus naive average

100%

collapse avoidance

fixed baseline scenarios recovered

ONE-CLICK JUDGE PROOF

Inject emergency -> block unsafe proposal -> admit safe recovery -> inspect measured chemistry

Keep the food-control loop local.

Three continuity layers separate transport, intelligence, and power claims.



NO WI-FI

DECT NR+ carries the tank-to-gateway hop.

NO CLOUD

Self-hosted Gemma and deterministic rules remain local.

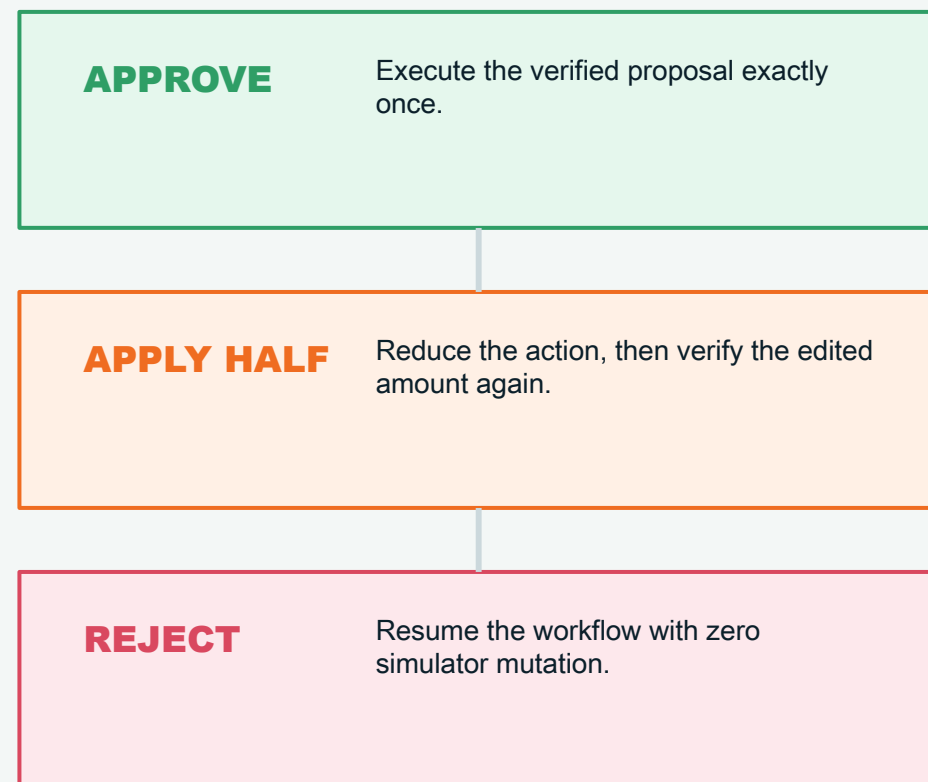
NO GRID

Solar + battery is the next measured autonomy proof.

Risky actions stop for a human decision.

The producer sees the proposed change, safety result, and current chemistry before choosing what happens next.

No irreversible action runs silently.



One endpoint. Two deployment paths.

Agent and verifier code stay unchanged when the inference host changes.

PROVEN NOW

CPU edge runtime

Gemma 4 E2B
llama.cpp QAT Q4
private 8 GB host
separate Metal evidence



GEMMA_ENDPOINT

OpenAI-compatible model boundary



PORTABLE AMD PATH

GPU promotion profile

AMD ROCm
vLLM serving
same endpoint contract
GPU evidence not claimed

The deterministic verifier remains authoritative on either host.

Protect protein production where connectivity cannot be assumed.

A verifier-gated, producer-controlled platform for rural farms, cooperatives, and food-security programs.

PRIVATE FIELD LINK	Two physical nRF9151 radios
LOCAL INTELLIGENCE	Self-hosted Gemma 4
EXECUTABLE SAFETY	Deterministic verifier + RLVR

proteinloop.dev-vb.lat | Best Unicorn

